General Mathematics (Further Maths 2020–22 set) · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

Exam 1 · multiple choice	Exam 2 · written response	State average on these questions
40 marks	**61 marks**	**55 / 101 (54%)**
reading 15 min · writing 1 h 30 min	reading 15 min · writing 1 h 30 min	a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

These are real VCAA questions from the 2020–2022 Further Mathematics exams, the subject General Maths replaced in 2023. Only the core (data analysis, recursion and finance) and the Matrices and Networks modules are used, and questions on content that isn't in the current General Maths study design (solving simultaneous equations with matrices) are left out. They were ranked by how Further Maths students actually scored on them within each area of study, and this paper takes the middle band.

Sources

- 2020: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2021: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2022: Exam 1 (PDF) · report · Exam 2 (PDF) · report

Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. Further Maths questions have five options (A–E).

Data analysis

Question 1 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 3 · open in the official exam PDF (p. 5) ↗

Shape of the same distribution from its five-number summary

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 2 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 7 · open in the official exam PDF (p. 6) ↗

How many of five insect variables are discrete numerical

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 3 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 10 · open in the official exam PDF (p. 8) ↗

Types of the variables frequency of nightmares (low, high) and snores (no, yes)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 4 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 12 · open in the official exam PDF (p. 8) ↗

Percentage of snorers with a high frequency of nightmares

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 5 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 16 · open in the official exam PDF (p. 10) ↗

Missing December long-term mean rainfall from seasonal indices

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 6 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 20 · open in the official exam PDF (p. 12) ↗

Five-mean smoothed passengers for March, given the January–May total

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 7 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 1 · open in the official exam PDF (p. 4) ↗

Types of the variables age (under 55, 55 and over) and preferred travel destination

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 8 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 8 · open in the official exam PDF (p. 9) ↗

Which of three participants reach a standardised score of at least 1.80

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 9 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 9 · open in the official exam PDF (p. 9) ↗

Mean and SD of heights from two given percentages

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 10 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 10 · open in the official exam PDF (p. 9) ↗

Which earlier day a least squares line predicts day 10's walking time will match

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 11 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 15 · open in the official exam PDF (p. 13) ↗

Seasonal index for summer from three years of quarterly visitor numbers

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 12 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 3 · open in the official exam PDF (p. 5) ↗

Which value could be the third quartile of the skull widths

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 13 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 7 · open in the official exam PDF (p. 7) ↗

Increase in $\log_{10}$(spleen weight) per one-month increase in age

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 14 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 8 · open in the official exam PDF (p. 7) ↗

Predicted spleen weight of a 30-month-old seal from the $\log_{10}$ model

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 15 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 10 · open in the official exam PDF (p. 8) ↗

Percentage of A-grade students who were taught individually

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 16 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 12 · open in the official exam PDF (p. 10) ↗

Least squares line for body length against head length, from a scatterplot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Recursion and financial modelling

Question 17 · 1 mark

2020 VCAA Further Maths Exam 1 · Core · Question 28 · open in the official exam PDF (p. 15) ↗

Which effective interest rate for a nominal 8% p.a. is wrong

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 18 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 19 · open in the official exam PDF (p. 14) ↗

Number of years an annuity pays $44 970.55 a year

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 19 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 20 · open in the official exam PDF (p. 15) ↗

Value whose 10% is the third year's reducing balance depreciation of a $72 000 boat

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 20 · 1 mark

2021 VCAA Further Maths Exam 1 · Core · Question 22 · open in the official exam PDF (p. 15) ↗

Monthly addition needed for $12 000 at 2.8% p.a. to reach $25 000 in five years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 21 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 19 · open in the official exam PDF (p. 12) ↗

Time to repay the same loan

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 22 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 20 · open in the official exam PDF (p. 12) ↗

Unit cost rate for equipment losing $5475 a year at 10 hours a day

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 23 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 22 · open in the official exam PDF (p. 13) ↗

Monthly recurrence relation matching the yearly rule $T_n = 6000 \times 1.003^{12n}$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 24 · 1 mark

2022 VCAA Further Maths Exam 1 · Core · Question 23 · open in the official exam PDF (p. 14) ↗

Simple interest recurrence relation that ends at the same value as $4000 at 3.88% compound over five years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Matrices

Question 25 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 3 · open in the official exam PDF (p. 18) ↗

Smallest power n of a permutation matrix that returns a column of letters to its original order

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 26 · 1 mark

2020 VCAA Further Maths Exam 1 · Matrices module · Question 9 · open in the official exam PDF (p. 23) ↗

Which player is sure to rank first on one- and two-step dominances by winning their remaining match

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 27 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 2 · open in the official exam PDF (p. 18) ↗

Transition matrix for workers buying lunch at a deli or a cafe

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 28 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 4 · open in the official exam PDF (p. 19) ↗

Permutation matrix that reorders R, A, M, O, N into N, O, R, M, A

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 29 · 1 mark

2021 VCAA Further Maths Exam 1 · Matrices module · Question 6 · open in the official exam PDF (p. 20) ↗

Missing label w on a partly labelled transition diagram for four classes

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 30 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 2 · open in the official exam PDF (p. 16) ↗

Matrix calculation for Sunday prices that are 10% higher

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 31 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 3 · open in the official exam PDF (p. 17) ↗

Transition matrix matching an incomplete transition diagram for swim sessions

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 32 · 1 mark

2022 VCAA Further Maths Exam 1 · Matrices module · Question 5 · open in the official exam PDF (p. 18) ↗

Which product of matrices of given orders could have an inverse

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Networks and decision mathematics

Question 33 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 4 · open in the official exam PDF (p. 26) ↗

Number of vertices that can be reached from S in a directed graph of one-way streets

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 34 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 5 · open in the official exam PDF (p. 26) ↗

Smallest value of an unknown edge x, given the minimum spanning tree length

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 35 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 6 · open in the official exam PDF (p. 27) ↗

Minimum completion time of an activity network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 36 · 1 mark

2020 VCAA Further Maths Exam 1 · Networks module · Question 9 · open in the official exam PDF (p. 29) ↗

How many of five labelled cuts have capacity equal to the maximum flow

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 37 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 6 · open in the official exam PDF (p. 25) ↗

Largest duration x for activity E that keeps an 18-hour completion time

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 38 · 1 mark

2021 VCAA Further Maths Exam 1 · Networks module · Question 8 · open in the official exam PDF (p. 26) ↗

Minimum length of road to clear, for different values of two unknown edges

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 39 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 5 · open in the official exam PDF (p. 23) ↗

Which statement about a connected graph with five vertices and four edges is false

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 40 · 1 mark

2022 VCAA Further Maths Exam 1 · Networks module · Question 6 · open in the official exam PDF (p. 23) ↗

Earliest start time of activity G in a network with a dummy activity

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Exam 2 · written response (61 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Data analysis

Question 1 (5 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 3 · open in the official exam PDF (p. 6) ↗

BMI for three neck-size groups of men, given as five-number summaries and parallel boxplots

a. Percentage of the 250 men with a below-average neck size (1 mark)

b. IQR of BMI for men with an average neck size (1 mark)

c. Number of the 250 men with a BMI of 30 or more (1 mark)

d. Do the boxplots support an association between BMI and neck size? Quote a statistic (2 marks)

Question 2 (6 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 3 · open in the official exam PDF (p. 6) ↗

Women's Olympic 100 m freestyle winning times since 1956, with a least squares trend line ($r^2 = 0.8794$)

a. Explanatory variable (1 mark)

b. Correlation coefficient, to three decimal places (1 mark)

c. Average decrease in winning time per year (1 mark)

d. Residual for 2000 (predicted 54.10 s, actual 53.83 s) (1 mark)

e.i. Show the predicted winning time for 2032 is 49.252 s (1 mark)

e.ii. Assumption made in that prediction (1 mark)

Question 3 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 5 · open in the official exam PDF (p. 10) ↗

The yearly gap between women's and men's winning times since 1912, linearised with a reciprocal transformation

a. Fit a least squares line to 1/difference against year, to four significant figures (2 marks)

b. Predicted difference in 2032, to one decimal place (1 mark)

Question 4 (6 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 1 · open in the official exam PDF (p. 2) ↗

A back-to-back stem plot of daily maximum wind speed in April and November: median, percentages, outliers and a histogram

a.i. Median April wind speed (1 mark)

a.ii. Percentage of April days with wind speed below 25 km/h (1 mark)

b. Show that 59 and 70 km/h are outliers for November (Q1 = 22, Q3 = 35) (2 marks)

c. Histogram of November wind speeds with class width 5 starting at 15 (2 marks)

Question 5 (4 marks)

2022 VCAA Further Maths Exam 2 · Core · Question 5 · open in the official exam PDF (p. 10) ↗

A 36-month rainfall time series: features, median smoothing and nine-mean smoothing

a. Two features of the plot apart from irregular fluctuations (1 mark)

b. Five-median smoothed rainfall for month 20 (1 mark)

c.i. Nine-mean smoothed rainfall for month 7, to one decimal place (1 mark)

c.ii. Number of points in the smoothed plot if nine-mean smoothing is applied to all 36 months (1 mark)

Recursion and financial modelling

Question 6 (3 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 8 · open in the official exam PDF (p. 15) ↗

Amortisation table for a $320 000 loan at 3.6% p.a. with monthly repayments of $1600

a.i. Principal reduction for payment 3 (1 mark)

a.ii. Balance after payment 4, to the nearest cent (1 mark)

b. Recurrence relation for the balance S_n (1 mark)

Question 7 (3 marks)

2020 VCAA Further Maths Exam 2 · Core · Question 10 · open in the official exam PDF (p. 17) ↗

A $500 000 annuity modelled by $A_{n+1} = kA_n - 2000$

a. Balance after two months when k = 1.0024 (1 mark)

b. Annual interest rate when k = 1.0024 (1 mark)

c. Value of k for which the annuity is a simple perpetuity (1 mark)

Question 8 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 6 · open in the official exam PDF (p. 12) ↗

A $420 000 perpetuity at 5.4% p.a. paying $1890 a month

a. Total received in one year (1 mark)

b. Value of the perpetuity after one year of payments (1 mark)

c. Recurrence relation for the value S_n (1 mark)

Question 9 (3 marks)

2021 VCAA Further Maths Exam 2 · Core · Question 7 · open in the official exam PDF (p. 12) ↗

A $12 000 coffee machine depreciated by unit cost at $0.05 a cup: $M_0 = 12\,000$, $M_{n+1} = M_n - 1440$

a. Cups of coffee made per year (1 mark)

b. Equivalent annual flat rate percentage (1 mark)

c. Rule for the value after n cups (1 mark)

Matrices

Question 10 (6 marks)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 1 · open in the official exam PDF (p. 20) ↗

Shopper numbers at three shopping centres and the proportions in each shopping area, combined by matrix multiplication

a. Order of the row matrix V (1 mark)

b. Total Grandmall shoppers if 700 are in its merchandise area (25%) (1 mark)

c.i. Two missing elements of Q = P × V (1 mark)

c.ii. Meaning of element q_{23} (1 mark)

d. Matrix calculation, using the average-spend matrix, that gives the total of $9663.20 (1 mark)

e. Matrix K that applies changes of +5%, −15% and −1% to the average spends (1 mark)

Question 11 (2 marks)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 4 · open in the official exam PDF (p. 25) ↗

A model with additions, $R_{n+1} = TR_n + B$, for shoppers at the three centres, given R_2

a. Expected Westmall shoppers in week 3 (1 mark)

b. Expected Westmall shoppers in week 1 (1 mark)

Question 12 (3 marks)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 3 · open in the official exam PDF (p. 18) ↗

Students switching each day between beginner, intermediate and advanced ski runs

a. Number of skiers expected to choose the same run on 17 July as on 16 July (1 mark)

b. Percentage of 17 July's advanced skiers who were also advanced on 16 July (1 mark)

c. Maximum number expected on the intermediate run on any one day (1 mark)

Networks and decision mathematics

Question 13 (3 marks)

2020 VCAA Further Maths Exam 2 · Networks module · Question 4 · open in the official exam PDF (p. 30) ↗

A directed network of track capacities, in people per minute, from S to O

a. Number of different routes from S to O (1 mark)

b. Capacity of Cut 1 (1 mark)

c. Maximum flow from S to O (1 mark)

Question 14 (3 marks)

2021 VCAA Further Maths Exam 2 · Networks module · Question 3 · open in the official exam PDF (p. 24) ↗

A directed road network with capacities in vehicles per hour, from the entrance to the exit of a town

a. Maximum flow from entrance to exit (1 mark)

b.i. The one road whose capacity should be increased (1 mark)

b.ii. Minimum new capacity of that road to maximise the flow (1 mark)

Question 15 (3 marks)

2021 VCAA Further Maths Exam 2 · Networks module · Question 4 · open in the official exam PDF (p. 25) ↗

A 13-activity roadworks network, then crashing selected activities at a weekly cost

a. Earliest start time of activity K (1 mark)

b. Number of activities with zero float (1 mark)

c. Minimum cost of reducing the completion time to 16 weeks (1 mark)

Question 16 (5 marks)

2022 VCAA Further Maths Exam 2 · Networks module · Question 2 · open in the official exam PDF (p. 23) ↗

A 10-activity project network for building a new cabin, then adding the cabin to the road network from an adjacency matrix

a. Number of activities with two immediate predecessors (1 mark)

b. Minimum completion time (1 mark)

c. Two activities to reduce (by 2 weeks and 1 week) to cut the completion time by three weeks (1 mark)

d. Add the new cabin H and its roads using the adjacency matrix (2 marks)

Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Exam 1 · multiple choice

✓	Q	Ans	Why	How Victoria went
☐	1	**C**	Q3 − median (150) is much bigger than median − Q1 (80) and the maximum is far out, so it is positively skewed. Upper fence = 300 + 1.5 × 230 = 645, and the maximum of 1380 is beyond it: positively skewed with outliers. E picked the wrong direction.	59% correct A 2% · B 7% · C 59% · D 5% · E 27%
☐	2	**A**	Only the number of wings is counted. Colour and species are categorical; length and weight are measured (continuous).	69% correct A 69% · B 15% · C 14% · D 1% · E 0%
☐	3	**A**	Low/high has a natural order (ordinal); no/yes doesn't (nominal). Many called both ordinal.	57% correct A 57% · B 5% · C 3% · D 24% · E 11%
☐	4	**B**	Use the snorers' column only: 12 ÷ 70 = 17.1%. A (7.5%) divided by the grand total instead.	70% correct A 21% · B 70% · C 4% · D 6% · E 0%
☐	5	**D**	Each month's mean ÷ its index gives the same annual mean, 51.9 ÷ 0.728 ≈ 71.3. December = 1.072 × 71.3 ≈ 76.4 mm.	67% correct A 5% · B 7% · C 13% · D 67% · E 8%
☐	6	**D**	The five-mean centred on March is the January–May mean: 2 160 000 ÷ 5 = 432 000.	57% correct A 17% · B 12% · C 8% · D 57% · E 6%
☐	7	**A**	Age recorded in two groups is categorical, and so is destination. C treated grouped age as numerical.	70% correct A 70% · B 3% · C 21% · D 4% · E 2%
☐	8	**C**	z = (score − 69.5) ÷ 6.5: Amy 1.85, Brian 1.69, Cherie 1.92. Only Brian falls short.	67% correct A 3% · B 12% · C 67% · D 3% · E 15%
☐	9	**C**	16% above 160 means 160 = μ + σ; 2.5% below 115 means 115 = μ − 2σ. Subtracting: 3σ = 45, so σ = 15 and μ = 145.	71% correct A 8% · B 10% · C 71% · D 7% · E 4%
☐	10	**B**	Fit time against day on CAS: time = 44 − 1 × day. Day 10 gives 34 minutes, the same as day 4.	71% correct A 7% · B 71% · C 13% · D 5% · E 4%
☐	11	**C**	Divide each summer value by its year's quarterly average (1.0916, 1.0965, 1.0770) and average the three: about 1.088.	59% correct A 10% · B 8% · C 59% · D 15% · E 6%
☐	12	**D**	A quarter of 46 is 11.5. Counting down from the top, the bars from 58 mm up hold 10 possums and adding the 57–58 mm bar passes 11.5, so Q3 is in 57–58: 57.7.	55% correct A 4% · B 5% · C 13% · D 55% · E 23%
☐	13	**A**	That is the slope of the transformed line: 0.009434.	68% correct A 68% · B 5% · C 10% · D 15% · E 2%
☐	14	**D**	$\log_{10}$(weight) = 2.698 + 0.009434 × 30 = 2.981, so weight = 10^2.981 ≈ 957 g. A forgot to undo the log.	56% correct A 28% · B 7% · C 6% · D 56% · E 3%

☐	15	**C**	Use the A row only: 10 ÷ 28 ≈ 36%. A (9%) divided by the individual column total instead.	56% correct A 34% · B 2% · C 56% · D 7% · E 1%
☐	16	**B**	Head length is explanatory, so body length = a + b × head length. Two points on the line give a slope of about 7 and an intercept of about −40.	58% correct A 4% · B 58% · C 13% · D 13% · E 12%
☐	17	**C**	Fortnightly: $(1 + 0.08/26)^{26} - 1 \approx 8.32\%$, not 8.31%. The other four are correct.	58% correct A 13% · B 5% · C 58% · D 9% · E 14%
☐	18	**B**	The interest rate is 20 000 ÷ 500 000 = 4% p.a. Finance Solver (or iterating the recurrence) gives N = 15 payments.	44% correct A 24% · B 44% · C 10% · D 15% · E 6%
☐	19	**C**	Third-year depreciation is 10% of the value at the start of year 3: $72\,000 \times 0.9^2 = \$58\,320$. B is the value after three years.	44% correct A 3% · B 40% · C 44% · D 9% · E 4%
☐	20	**B**	Finance Solver: N = 60, I% = 2.8, PV = −12 000, FV = 25 000, monthly: PMT ≈ −174.11, so $174.11.	56% correct A 5% · B 56% · C 10% · D 23% · E 5%
☐	21	**A**	R = 1.003 means 3.6% p.a. Finance Solver with PV = 400 000 and PMT = −2024 gives N ≈ 300 months: 25 years with a slightly different final payment.	47% correct A 47% · B 15% · C 12% · D 14% · E 10%
☐	22	**A**	10 × 365 = 3650 hours a year, and 5475 ÷ 3650 = $1.50 per hour. D gave the same number per day.	65% correct A 65% · B 6% · C 8% · D 15% · E 5%
☐	23	**C**	1.003^{12n} means 12n months growing by a factor of 1.003 each: $R_0 = 6000$, $R_{n+1} = 1.003R_n$. E multiplied the rate by 12.	58% correct A 4% · B 4% · C 58% · D 11% · E 23%
☐	24	**D**	Li ends with $4000 \times 1.0388^5 = \$4838.60$. Check each option's $J_0 + 5d$: only D gives 3500 + 5 × 267.72 = 4838.60.	49% correct A 8% · B 15% · C 10% · D 49% · E 18%
☐	25	**D**	P moves four of the letters around a cycle and fixes one, so P^4 is the identity: n = 4. A (n = 1) didn't check the result.	64% correct A 19% · B 7% · C 4% · D 64% · E 5%
☐	26	**B**	Fill in each possible result and add the one-step and two-step dominances. If Brie beats Andy she scores 8 or 9, ahead of everyone else whichever way the other match goes. No one else can guarantee first place.	63% correct A 5% · B 63% · C 21% · D 6% · E 5%
☐	27	**D**	Columns are "this Friday" and add to 1: deli stays 0.65 and moves 0.35; cafe stays 0.55 and moves 0.45 to the deli.	73% correct A 2% · B 7% · C 5% · D 73% · E 13%
☐	28	**E**	Each row has a 1 in the column of the letter it picks up: row 1 takes N (5th), row 2 O (4th), row 3 R (1st), row 4 M (3rd), row 5 A (2nd).	73% correct A 4% · B 8% · C 8% · D 7% · E 73%
☐	29	**B**	Match the diagram's edges to the matrix: the pair joined by 20% both ways can only be A and C, and 25% can only be B to D. That places every class, and w is the transition from A to D: 15%.	66% correct A 17% · B 66% · C 7% · D 5% · E 4%
☐	30	**D**	Increasing by 10% multiplies every element by 1.1: 1.1B.	68% correct A 2% · B 24% · C 6% · D 68% · E 0%
☐	31	**C**	Use the labels you have and fill the gaps so every column adds to 1. C is the only option that matches the diagram with columns adding to 1. E puts zeros where the diagram has arrows.	71% correct A 1% · B 3% · C 71% · D 2% · E 22%

☐	32	**D**	Only square matrices have inverses. GF is 3 × 2 times 2 × 3 = 3 × 3; the other products are not square.	71% correct A 13% · B 6% · C 5% · D 71% · E 4%
☐	33	**C**	Follow the arrows: S reaches T and V directly and then W through V. The edges at U and X point the wrong way, so three vertices. The popular wrong answer (2) missed W.	39% correct A 1% · B 58% · C 39% · D 2% · E 0%
☐	34	**D**	Prim's algorithm without x gives 5 + 6 + 7 + 9 + 6 + 5 + 8 + 7 = 53, using the 9 m edge from G to I. If x were less than 9 it would replace that edge and the total would drop, so x ≥ 9.	56% correct A 13% · B 13% · C 12% · D 56% · E 6%
☐	35	**E**	Forward scan: the longest path is B–E–I–J–M = 7 + 4 + 3 + 3 + 5 = 22 days.	61% correct A 16% · B 6% · C 11% · D 5% · E 61%
☐	36	**C**	Find each cut's capacity, counting only edges that cross from the source side to the sink side. The smallest capacity is the maximum flow, and three of the cuts share that value.	60% correct A 8% · B 18% · C 60% · D 9% · E 4%
☐	37	**B**	Two paths run through E: B–E–J allows x ≤ 6, but C–G–F–E–J (4 + 3 + 1 + x + 7 ≤ 18) allows only x ≤ 3. So x is at most 3. E (6) checked only one path.	46% correct A 7% · B 46% · C 6% · D 8% · E 31%
☐	38	**B**	Run Prim's algorithm for each option. With x = 50 and y = 60 the minimum spanning tree is 361 km. The other pairs give 356, 358 or 363 km, not the totals stated.	48% correct A 16% · B 48% · C 13% · D 16% · E 6%
☐	39	**E**	A connected graph with one fewer edge than vertices is a tree, so it can't contain a cycle. It is still planar, bipartite and contains paths.	54% correct A 9% · B 8% · C 21% · D 6% · E 54%
☐	40	**C**	G needs both D (6 + 5 = 11) and, through the dummy, B and E (5 + 7 = 12). The later of these is 12.	48% correct A 20% · B 25% · C 48% · D 4% · E 2%

Multiple-choice score: ☐ **/ 40** **(state average 23.9)**

Exam 2 · written response

Question 1 (5 marks) · state average 3 / 5 (60%)

2020 VCAA Further Maths Exam 2 · Core · Question 3

a. Percentage of the 250 men with a below-average neck size

☐ 20%

Answer / notes: 50 ÷ 250 = 20%.

State average 0.9 / 1 (90%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: ☐ **/ 1**

b. IQR of BMI for men with an average neck size

☐ 2.6

Answer / notes: 26.0 − 23.4 = 2.6.

State average 0.9 / 1 (90%); 89% scored full marks. Mark distribution: 0: 11% · 1: 89%

Your mark: ☐ **/ 1**

c. Number of the 250 men with a BMI of 30 or more

- [ ] 23

Answer / notes: For above-average necks Q3 = 29.95, so a quarter of 76 (19 men) have BMI ≥ 30, as values are rounded to one decimal place. Add the four average-group outliers above 30: 23. Counting the above-average group's two outliers again gave 25.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

Your mark: ☐ **/ 1**

d. Do the boxplots support an association between BMI and neck size? Quote a statistic

- [ ] Yes: median BMI increases as neck size increases
- [ ] Quotes the medians 21.6, 24.6 and 28.1 (or the IQRs)

Answer / notes: Yes. Median BMI rises from 21.6 to 24.6 to 28.1 as neck size goes from below average to above average. Stick to one statistic and quote its values from the table.

State average 1.0 / 2 (50%); 36% scored full marks. Mark distribution: 0: 38% · 1: 26% · 2: 36%

Your mark: ☐ **/ 2**

Question 2 (6 marks) · state average 3.2 / 6 (53%)

2021 VCAA Further Maths Exam 2 · Core · Question 3

a. Explanatory variable

- [ ] year

Answer / notes: Year.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: ☐ **/ 1**

b. Correlation coefficient, to three decimal places

- [ ] −0.938

Answer / notes: r = −√0.8794 = −0.938: negative because the trend is decreasing. Most left off the sign.

State average 0.2 / 1 (20%); 19% scored full marks. Mark distribution: 0: 81% · 1: 19%

Your mark: ☐ **/ 1**

c. Average decrease in winning time per year

- [ ] 0.1515 seconds per year

Answer / notes: The slope tells you directly: 0.1515 s per year. "Write down" means no calculation is needed.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: ☐ **/ 1**

d. Residual for 2000 (predicted 54.10 s, actual 53.83 s)

- [ ] −0.27

Answer / notes: Actual − predicted = 53.83 − 54.10 = −0.27 s.

State average 0.7 / 1 (70%); 68% scored full marks. Mark distribution: 0: 32% · 1: 68%

Your mark: ☐ **/ 1**

e.i. Show the predicted winning time for 2032 is 49.252 s

- [ ] 357.1 − 0.1515 × 2032 = 49.252

Answer / notes: Substitute and show it: 357.1 − 0.1515 × 2032 = 49.252.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: ☐ **/ 1**

e.ii. Assumption made in that prediction

- [ ] The same decreasing linear trend continues into the future

Answer / notes: That the linear trend continues beyond the data. Just saying “extrapolation” wasn’t enough.

State average 0.2 / 1 (20%); 18% scored full marks. Mark distribution: 0: 82% · 1: 18%

Your mark: ____ **/ 1**

Question 3 (3 marks) · state average 0.7 / 3 (23%)

2021 VCAA Further Maths Exam 2 · Core · Question 5

a. Fit a least squares line to 1/difference against year, to four significant figures

- [ ] Uses 1/difference as the response
- [ ] 1/difference = −2.234 + 0.001209 × year

Answer / notes: CAS on (year, 1/difference) gives 1/difference = −2.234 + 0.001209 × year. Rounding and naming the variables tripped many up.

State average 0.5 / 2 (25%); 16% scored full marks. Mark distribution: 0: 65% · 1: 19% · 2: 16%

Your mark: ____ **/ 2**

b. Predicted difference in 2032, to one decimal place

- [ ] 4.5 seconds

Answer / notes: −2.234 + 0.001209 × 2032 ≈ 0.222, so difference ≈ 1 ÷ 0.222 ≈ 4.5 s. Many forgot to take the reciprocal and wrote 0.2.

State average 0.2 / 1 (20%); 14% scored full marks. Mark distribution: 0: 86% · 1: 14%

Your mark: ____ **/ 1**

Question 4 (6 marks) · state average 4.5 / 6 (75%)

2022 VCAA Further Maths Exam 2 · Core · Question 1

a.i. Median April wind speed

- [ ] 20 km/h

Answer / notes: With 30 values, the median is the mean of the 15th and 16th: both 20 km/h.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: ____ **/ 1**

a.ii. Percentage of April days with wind speed below 25 km/h

- [ ] 60%

Answer / notes: 18 of the 30 April days are below 25 km/h: 60%.

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: ____ **/ 1**

b. Show that 59 and 70 km/h are outliers for November (Q1 = 22, Q3 = 35)

- [ ] Upper fence = 35 + 1.5 × 13 = 54.5
- [ ] 59 and 70 are both above 54.5, so they are outliers

Answer / notes: IQR = 13, so the upper fence is 54.5. Both values are above it. Finding the fence without saying why that makes them outliers lost a mark.

State average 1.5 / 2 (75%); 64% scored full marks. Mark distribution: 0: 19% · 1: 16% · 2: 64%

Your mark: ____ **/ 2**

c. Histogram of November wind speeds with class width 5 starting at 15

- [ ] Correct column heights 4, 7, 5, 5, 3, 1, 2, 1, 1, 0, 0, 1
- [ ] Columns drawn from 15 to 75 with no gaps except the empty classes

Answer / notes: Count each class from the stem plot: 15–20: 4, 20–25: 7, 25–30: 5, 30–35: 5, 35–40: 3, 40–45: 1, 45–50: 2, 50–55: 1, 55–60: 1, then two empty classes and 1 at 70–75. Missing a column or misplacing the 70 was common.

State average 1.3 / 2 (65%); 61% scored full marks. Mark distribution: 0: 28% · 1: 11% · 2: 61%

Your mark: ☐ **/ 2**

Question 5 (4 marks) · state average 2 / 4 (50%)

2022 VCAA Further Maths Exam 2 · Core · Question 5

a. Two features of the plot apart from irregular fluctuations

- [ ] Decreasing trend and seasonality

Answer / notes: A decreasing trend and seasonality. You needed both. Structural change was a common wrong guess.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ☐ **/ 1**

b. Five-median smoothed rainfall for month 20

- [ ] 100 mm

Answer / notes: Take months 18 to 22 off the plot and find the middle value: 100 mm. Some calculated a mean.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: ☐ **/ 1**

c.i. Nine-mean smoothed rainfall for month 7, to one decimal place

- [ ] 133.5

Answer / notes: Mean of months 3 to 11: 1201.2 ÷ 9 = 133.5 mm.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: ☐ **/ 1**

c.ii. Number of points in the smoothed plot if nine-mean smoothing is applied to all 36 months

- [ ] 28

Answer / notes: Four months are lost at each end, so 36 − 8 = 28. 4 was a common slip.

State average 0.4 / 1 (40%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

Your mark: ☐ **/ 1**

Question 6 (3 marks) · state average 1.5 / 3 (50%)

2020 VCAA Further Maths Exam 2 · Core · Question 8

a.i. Principal reduction for payment 3

- [ ] $643.85

Answer / notes: 1600 − 956.15 = $643.85.

State average 0.8 / 1 (80%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: ☐ **/ 1**

a.ii. Balance after payment 4, to the nearest cent

- [ ] $317 428.45

Answer / notes: Interest = 0.003 × 318 074.23 = 954.22, so the balance = 318 074.23 + 954.22 − 1600 = $317 428.45.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

Your mark: ☐ **/ 1**

b. Recurrence relation for the balance S_n

- [ ] $S_0 = 320\,000$, $S_{n+1} = 1.003S_n - 1600$

Answer / notes: 3.6% ÷ 12 = 0.3% a month, so $S_0 = 320\,000$, $S_{n+1} = 1.003S_n - 1600$. Wrong factors seen: 0.003, 0.997 and 1.036.

State average 0.3 / 1 (30%); 30% scored full marks. Mark distribution: 0: 70% · 1: 30%

Your mark: ☐ **/ 1**

Question 7 (3 marks) · state average 1.5 / 3 (50%)

2020 VCAA Further Maths Exam 2 · Core · Question 10

a. Balance after two months when k = 1.0024

- [ ] $498 398.08

Answer / notes: $A_1 = 499\,200$, $A_2 = 1.0024 \times 499\,200 - 2000 = \$498\,398.08$.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

b. Annual interest rate when k = 1.0024

- [ ] 2.88%

Answer / notes: 0.24% a month × 12 = 2.88% p.a.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: ☐ **/ 1**

c. Value of k for which the annuity is a simple perpetuity

- [ ] 1.004

Answer / notes: The interest must equal the payment: $500\,000(k - 1) = 2000$, so k = 1.004.

State average 0.4 / 1 (40%); 36% scored full marks. Mark distribution: 0: 64% · 1: 36%

Your mark: ☐ **/ 1**

Question 8 (3 marks) · state average 1.5 / 3 (50%)

2021 VCAA Further Maths Exam 2 · Core · Question 6

a. Total received in one year

- [ ] $22 680

Answer / notes: 12 × 1890 = $22 680.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: ☐ **/ 1**

b. Value of the perpetuity after one year of payments

- [ ] $420 000

Answer / notes: A perpetuity pays only the interest, so it stays at $420 000.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ☐ **/ 1**

c. Recurrence relation for the value S_n

- [ ] $S_0 = 420\,000$ and the factor 1.0045

Answer / notes: 5.4% ÷ 12 = 0.45% a month: $S_0 = 420\,000$, $S_{n+1} = 1.0045S_n - 1890$. 1.054 was a common error.

State average 0.4 / 1 (40%); 38% scored full marks. Mark distribution: 0: 62% · 1: 38%

Your mark: ☐ **/ 1**

Question 9 (3 marks) · state average 1.3 / 3 (43%)

2021 VCAA Further Maths Exam 2 · Core · Question 7

a. Cups of coffee made per year

☐ 28 800

Answer / notes: 1440 ÷ 0.05 = 28 800 cups.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: ____ **/ 1**

b. Equivalent annual flat rate percentage

☐ 12%

Answer / notes: 1440 ÷ 12 000 = 12%.

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: ____ **/ 1**

c. Rule for the value after n cups

☐ $M_n = 12\,000 - 0.05n$

Answer / notes: Here n counts cups, not years: $M_n = 12\,000 - 0.05n$. Few noticed the change.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: ____ **/ 1**

Question 10 (6 marks) · state average 3.9 / 6 (65%)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 1

a. Order of the row matrix V

☐ 1 × 3

Answer / notes: 1 × 3.

State average 1.0 / 1 (100%); 95% scored full marks. Mark distribution: 0: 5% · 1: 95%

Your mark: ____ **/ 1**

b. Total Grandmall shoppers if 700 are in its merchandise area (25%)

☐ 2800

Answer / notes: 700 ÷ 0.25 = 2800. Some just copied 2700.

State average 0.5 / 1 (50%); 45% scored full marks. Mark distribution: 0: 55% · 1: 45%

Your mark: ____ **/ 1**

c.i. Two missing elements of Q = P × V

☐ 1296 and 729

Answer / notes: Grandmall column: 0.48 × 2700 = 1296 and 0.27 × 2700 = 729.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: ____ **/ 1**

c.ii. Meaning of element q_{23}

☐ The number of shoppers (594) in the clothing area of Westmall

Answer / notes: Row 2 is clothing and column 3 is Westmall: 594 shoppers in Westmall's clothing area. Some read q_{32} instead.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: ____ **/ 1**

d. Matrix calculation, using the average-spend matrix, that gives the total of $9663.20

☐ $[135\ 143\ 131] \times A_{2019} = [9663.20]$

Answer / notes: A row of shopper numbers times the column of average spends: $[135\ 143\ 131] \times A_{2019}$. Getting the order wrong was common.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

e. Matrix K that applies changes of +5%, −15% and −1% to the average spends

☐ K = diagonal matrix with 1.05, 0.85, 0.99 on the leading diagonal

Answer / notes: K must be 3 × 3 so that $K \times A_{2019}$ is a column: a diagonal matrix with 1.05, 0.85 and 0.99 on the diagonal and zeros elsewhere. Few got this.

State average 0.2 / 1 (20%); 24% scored full marks. Mark distribution: 0: 76% · 1: 24%

Your mark: ☐ **/ 1**

Question 11 (2 marks) · state average 0.7 / 2 (35%)

2020 VCAA Further Maths Exam 2 · Matrices module · Question 4

a. Expected Westmall shoppers in week 3

☐ 237 966

Answer / notes: $R_3 = TR_2 + B$; its Westmall element is 237 966 (B removes 400 from Westmall each week). Say which element is the answer.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

b. Expected Westmall shoppers in week 1

☐ 241 000

Answer / notes: Work backwards: $R_1 = T^{-1}(R_2 - B)$, whose Westmall element is 241 000. Only 20% of students managed this.

State average 0.2 / 1 (20%); 20% scored full marks. Mark distribution: 0: 80% · 1: 20%

Your mark: ☐ **/ 1**

Question 12 (3 marks) · state average 1.1 / 3 (37%)

2022 VCAA Further Maths Exam 2 · Matrices module · Question 3

a. Number of skiers expected to choose the same run on 17 July as on 16 July

☐ 344

Answer / notes: 0.7 × 210 + 0.7 × 190 + 0.8 × 80 = 344.

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: ☐ **/ 1**

b. Percentage of 17 July's advanced skiers who were also advanced on 16 July

☐ 63%

Answer / notes: 0.8 × 80 = 64 stayed, out of 0.2 × 190 + 64 = 102 advanced skiers: 64 ÷ 102 ≈ 63%.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ☐ **/ 1**

c. Maximum number expected on the intermediate run on any one day

☐ 219

Answer / notes: Keep calculating S_n: intermediate peaks at 219.04 three days later, so 219. Many gave the long-term value instead.

State average 0.2 / 1 (20%); 19% scored full marks. Mark distribution: 0: 81% · 1: 19%

Your mark: ☐ **/ 1**

Question 13 (3 marks) · state average 1.3 / 3 (43%)

2020 VCAA Further Maths Exam 2 · Networks module · Question 4

a. Number of different routes from S to O

☐ 10

Answer / notes: List every route following the arrows: there are 10. 9 was a common miscount.

State average 0.3 / 1 (30%); 30% scored full marks. Mark distribution: 0: 70% · 1: 30%

Your mark: ☐ **/ 1**

b. Capacity of Cut 1

☐ 52

Answer / notes: Add the edges crossing from the S side to the O side: 20 + 12 + 20 = 52.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: ☐ **/ 1**

c. Maximum flow from S to O

☐ 50

Answer / notes: Find the minimum cut: its capacity is 50 people per minute, smaller than Cut 1.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: ☐ **/ 1**

Question 14 (3 marks) · state average 0.8 / 3 (27%)

2021 VCAA Further Maths Exam 2 · Networks module · Question 3

a. Maximum flow from entrance to exit

☐ 1330

Answer / notes: The minimum cut is 1330 vehicles per hour. Many didn't look for a minimum cut at all.

State average 0.3 / 1 (30%); 24% scored full marks. Mark distribution: 0: 76% · 1: 24%

Your mark: ☐ **/ 1**

b.i. The one road whose capacity should be increased

☐ A to D

Answer / notes: A→D (650) is in the minimum cut, and raising it lets more flow through.

State average 0.4 / 1 (40%); 38% scored full marks. Mark distribution: 0: 62% · 1: 38%

Your mark: ☐ **/ 1**

b.ii. Minimum new capacity of that road to maximise the flow

☐ 780

Answer / notes: Raising A→D increases the flow until another cut takes over at 1460 (the 620 and 840 roads into the exit side). That needs A→D at 780. Almost nobody got this.

State average 0.1 / 1 (10%); 6% scored full marks. Mark distribution: 0: 94% · 1: 6%

Your mark: ☐ **/ 1**

Question 15 (3 marks) · state average 0.8 / 3 (27%)

2021 VCAA Further Maths Exam 2 · Networks module · Question 4

a. Earliest start time of activity K

☐ 14

Answer / notes: K needs F (3 + 5 = 8) and J (6 + 5 + 3 = 14): 14 weeks.

State average 0.5 / 1 (50%); 46% scored full marks. Mark distribution: 0: 54% · 1: 46%

Your mark: ☐ **/ 1**

b. Number of activities with zero float

- [ ] 7

Answer / notes: Two critical paths of 19 weeks: A–D–G–L–M and A–E–H–L–M. That makes A, D, E, G, H, L and M critical: seven. Many found only one path.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

Your mark: ____ **/ 1**

c. Minimum cost of reducing the completion time to 16 weeks

- [ ] $380 000

Answer / notes: L is on both critical paths: cut it 2 weeks ($240 000). A–E–J–K (17 weeks) then also needs cutting, so cut A by 1 week ($140 000), which shortens every long path. Total $380 000.

State average 0.1 / 1 (10%); 9% scored full marks. Mark distribution: 0: 91% · 1: 9%

Your mark: ____ **/ 1**

Question 16 (5 marks) · state average 3.2 / 5 (64%)

2022 VCAA Further Maths Exam 2 · Networks module · Question 2

a. Number of activities with two immediate predecessors

- [ ] 3

Answer / notes: H and I both follow D and F, and J follows G and H: three. 2 was the common error.

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: ____ **/ 1**

b. Minimum completion time

- [ ] 30 weeks

Answer / notes: Two critical paths of 30 weeks: A–C–F–I and A–D–I.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: ____ **/ 1**

c. Two activities to reduce (by 2 weeks and 1 week) to cut the completion time by three weeks

- [ ] A reduced by 2 weeks and I reduced by 1 week

Answer / notes: A is on every path, so cutting it 2 weeks saves 2. Then I is on both 28-week critical paths; cutting it 1 week brings everything down to 27 (A–C–F–H–J and A–D–H–J are also 27).

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ____ **/ 1**

d. Add the new cabin H and its roads using the adjacency matrix

- [ ] Vertex H joined to D and to E
- [ ] A loop at H

Answer / notes: Row H has 1s under D, E and H: draw H with edges to D and E and a loop.

State average 1.8 / 2 (90%); 80% scored full marks. Mark distribution: 0: 7% · 1: 13% · 2: 80%

Your mark: ____ **/ 2**

Exam 1: ____ **/ 40** **Exam 2:** ____ **/ 61** **Total:** ____ **/ 101**

State average on these exact questions: Exam 1 23.9 / 40, Exam 2 31.0 / 61, total 54.9 / 101 (54%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.